

Symbolic Logic

Solutions to Practice Second Examination

Part A. Satisfying Formulas (30 points) Use the following domain and interpretation to determine whether the provided expression is true or false.

D: Humans in the following diagram

B(1): is bored.	l: larrie.
C(1): is a cab driver.	r: ralph.
D(1): is a dog trainer.	
H(1): is hirsute.	
W(1): is wary.	

1. True or False: "Everything is human."

For all x, x is human.

T, since everything in the domain is human (we cannot represent this explicitly in our symbols though).

2. True or False: "All bored cab drivers are wary."

For all x such that x is bored and x is a cab driver, x is wary.

$\forall x((Bx \ \& \ Cx) \rightarrow Hx)$

T, since there are no bored cab drivers.

3. True or False: If Larrie and Ralph are human, then both are hirsute.

If larrie and ralph are human, then they are both hirsute. Effectively, we represent this as both ralph and larrie being hirsute, which is true.

$(Hl \ \& \ Hr)$

T, since both are hirsute.

4. True or False: $\sim \forall x([(Dx \ \& \ Wx] \vee Bx)$

It is not the case that for every x, x is a wary dog trainer, or bored. Alternatively, there is some

x such that it's not the case x is a wary dog trainer and it's not the case x is bored. That is, there is some x such that x is not a bored wary dog trainer.

T, since it is not the case that everything is bored or a wary dog trainer; Ralph is a witness.

5. True or False: $\forall x(Wx \rightarrow \exists x(Bx))$

For all x, if x is wary, then something is bored.

T, since it is the case that everything is wary, and Larrie is also bored.

Part B. Provide an Interpretation (30 points)

5. Provide an interpretation in a small finite domain that satisfies the following set of sentences (i.e. makes them all true). **Be sure to provide a complete interpretation, include the domain and explain your predicates and constants.**

$\exists x(Ax \vee Bx),$		$\forall x(\sim Bx \vee Ax), \sim \exists x(Ax \& Bx)$	
T		T	T

	A		
	9	9	

D: the smiley faces above
A: has an "A" above it. B:
has a "B" above it.

(You do *not* need to provide any of the following in your answer on the exam.)

The first sentence is true in this interpretation because there is something that either has an "A" above it or a "B" above it (namely the first smiley face).

The second sentence is true in this interpretation because everything is either not "B" or is "A", and indeed, the thing that is "A" is also not "B".

The third sentence is true in the above interpretation if and only if there is nothing in the domain that has both an "A" and a "B" above it. There is no such smiley face. So this sentence is true.

Expansions: (You do *not* need to provide these on the exam.)

$\exists x(Ax \vee Bx)$

(Aa $\vee$ Ba) $\vee$ (Ab $\vee$
Bb) T T F **T** F F F

$\forall x(\sim Bx \vee Ax)$

($\sim$ Ba $\vee$ Aa) &
($\sim$ Bb $\vee$ Ab) T F T T **T**
T F T F

$\sim \exists x(Ax \ \& \ Bx)$

$\sim [(Aa \ \& \ Ba) \vee (Ab \ \& \ Bb)]$ **T** **T** **F** **F** **F** **F** **F**

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7. Provide an interpretation in a small finite domain that satisfies the following set of sentences (i.e. makes them all true). **Be sure to provide a complete interpretation, include the domain and explain your predicates and constants.**

$\exists xFx \rightarrow \exists yGy$,		$\sim \forall xFx \vee \exists yGy$	
	T		T

	F	G	
	9	9	

D: The smiley faces above.
F: has an “F” above it. G:
has a “G” above it.

The first sentence is true on this interpretation, and the second sentence is true.

(None of the rest of this solution is required for full credit.)

Expansions:

$\exists xFx \rightarrow \exists yGy$

$\sim (FavFb) \rightarrow (GavGb)$

F **T** **T** **F** **T** **F** **T** **T**

$\sim \forall xFx \vee \exists yGy$

$(Fa \ \& \ Fb) \vee (Ga \vee Gb)$

T **F** **F** **T** **F** **T** **T**

Part C. Derivations (40 points)

Provide derivations that show that the following arguments are valid. (You will receive points for starting a derivation in a promising direction, even if you do not finish it. You will receive more points for an incomplete derivation that is correct as far as it goes than for a “complete” derivation that misuses the rules.)

8. $\sim\forall xFx \rightarrow \sim\exists yGy$; $\forall xFx \vee \exists yGy$; SHOW $\forall xFx$.

1. $\sim\forall xFx \rightarrow \sim\exists yGy$
2. $\forall xFx \vee \exists yGy$
3. SHOW $\forall xFx$ UD
4. SHOW Fa ID
5. $\sim Fa$ AID
6. $\exists x\sim Fx$ E-I
7. $\sim\forall xFx$ 6, Quantifier DeMorgan's
8. $\exists yGy$ 2,7, DE
9. $\forall xFx$ 1,8, Modus Tollens
10. ! 7,9 !I

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11. $(\exists yGy \rightarrow \exists xFx); (\forall xFx \vee \exists yGy); \sim Fa$; SHOW $\exists xFx$.

1. $(\exists yGy \rightarrow \exists xFx)$

2. $(\forall xFx \vee \exists yGy)$

3. $\sim Fa$

4. SHOW $\exists xFx$ DD

5. $\exists x\sim Fx$ E-I

6. $\sim \forall xFx$ Quantifier DeMorgan's

7. $\exists yGy$ 2,6, DE

8. $\exists xFx$ 1,7, MP

